

Hello Professor,

Below is an update for my week 14 progress. In addition, I have pushed my updates to my github repo: <https://github.com/Woyoung21/LastMile-Sec>. For week 14, the main task was to begin writing my paper. Additionally, I spent a good portion of the week creating my pitch video to demo my product for the SFS Student AI Awards. I will provide the links to my pitch video onYouTube, as well as the overall compilation of submissions created by the AT team. In general, this update will be a bit shorter than my development updates.

Here is the link for my pitch video: https://youtu.be/L5sriZ7e6oQ?si=klfgX4J_ihzwVQ5U . This was my very first pitch video, and I had to create a few different versions. I found it very difficult to condense a semester-long project into only 5 minutes. Especially with the technical breadth of this project. My goal was to really hone in on the architecture and then demo the product via the UI I created. In hind sight, I wish I would've spent more time showing the UI, especially walking through the various steps used to remediate different vulnerabilities. However I had to submit, and I will take that knowledge to the next project.

Here is the video compilation from the AT department (my submission is the last one in this compilation): <https://youtu.be/6TNyaFzOKs?si=TVfGAfIXslyyXPv7> . As you can see from the compilation, there was a wide variety of projects that were submitted for this competition. In my opinion, entering these competitions is always worth it, just to see what my other colleagues are working on. It's a great reminder that our school has a deep and broad bench of extremely talented students who are adapting rapidly to the newest technology available, and applying it to problems they see in the world.

In addition to my pitch video submission, I began working on my final paper. My target length is roughly in the ~10-20 page range. Structurewise, I will begin by giving an executive summary of the project and then dive into a section by section overview of the technical details of the project. These sections will also include any technical hurdles I had to overcome. Additionally I will add a section about any potential future development ideas that I have. Given more time, I would most likely continue to dial in the machine learning, move all of my cloud gemini calls to a local instance of Gemma (Google's open source model:

<https://deepmind.google/models/gemma/gemma-4/>). Additionally I might implement some PII redaction functions during the ingest section to remove any client identifying information and work on overall security of the project environment. If money wasn't an issue, I would also increase my local compute resources to give these local models more resources to work with.

This is my update for this week, I will follow up next week with a rough draft of the final paper.